Phase Jump Ride-through of Current-limited Grid-forming (GFM) Inverters

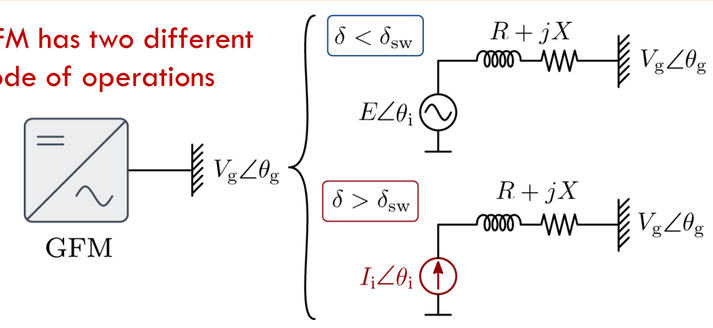
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Motivation

- Phase jumps can drive GFM inverters into current limiting almost instantaneously
- Under saturation, an unfavorable current direction can produce a severe active power dip, negative-power swing, and loss of synchronism
- Phase-jump ride-through therefore requires coordinated control of the limited-current magnitude, its angle, and the subsequent recovery

GFM Inverter Under Current Limiting (CL)

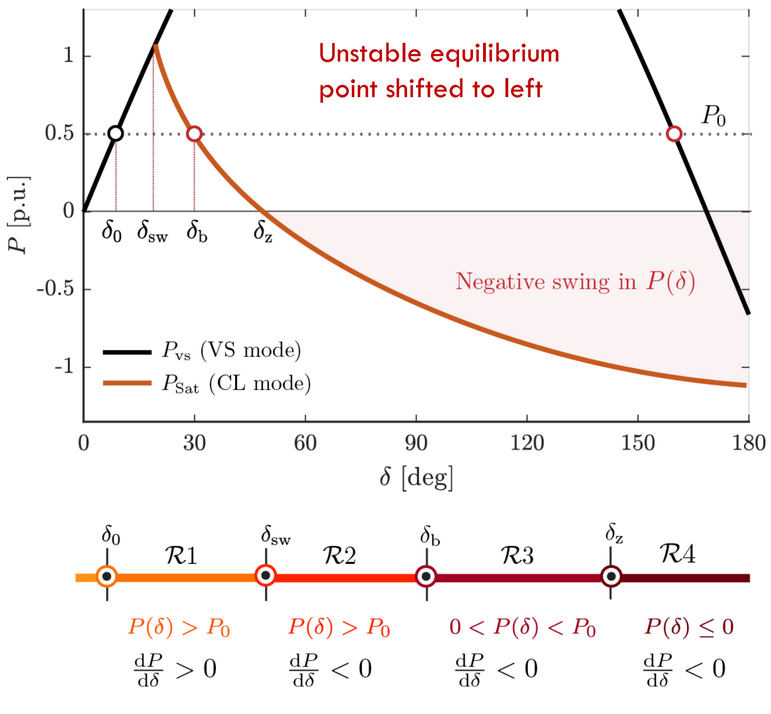
GFM has two different mode of operations



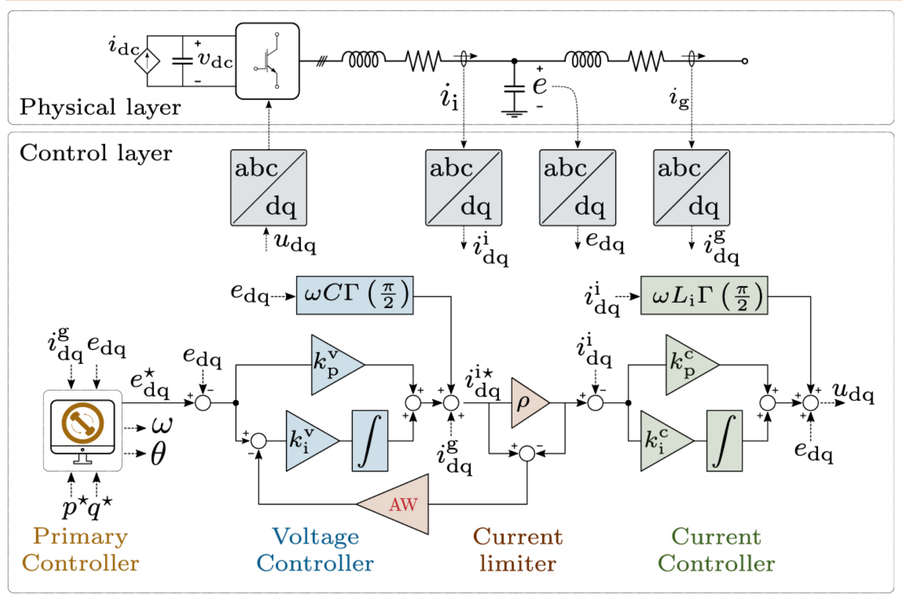
Power-angle Characteristics Under CL

As operating mode changes, power-angle characteristics change as well during current limiting

$$P(\delta) = \begin{cases} P_{vs}(\delta) = \dfrac{|E||V_g|}{X}\sin\delta, & 0 \le \delta < \delta_{sw}, \\ P_{Sat}(\delta) = |V_g| I_{lim}\cos\left[\Gamma(\delta) - \theta_{eff}^{Sat}(\delta)\right], & \delta_{sw} \le \delta \le \pi, \end{cases}$$



Generic GFM Control with Saturation-based Limiter



Why angle control matters

$$P_{Sat} = |V_g| I_{lim}\cos\phi, \qquad \phi(\delta) = \Gamma(\delta) - \theta_{eff}^{Sat}(\delta)$$

Current magnitude is fixed, but active power still depends on direction; ride-through therefore requires control of φ.

Current limiter: current reference saturator (SatLim)

$$I = \rho I^\star, \qquad \mu = \frac{1-\rho}{\rho} = \frac{|I^\star|}{I_{lim}} - 1 \ge 0, \qquad I^\star - I = \mu I$$

$\mu = 0$ at limiter entry; larger μ means severe fault

SatLim with scaler anti-windup (AW)

From network: $E - V_g = IZ_\ell$

From GFM control: $E = E^\star - k_w(I^\star - I)$

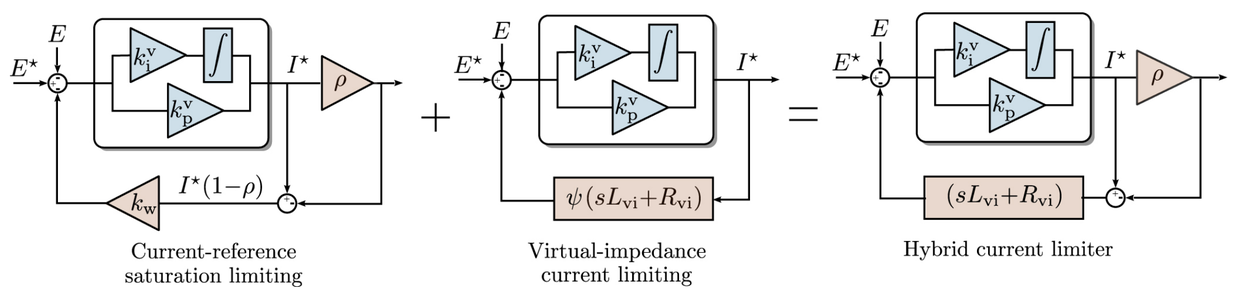
$$E^\star - V_g = I\underbrace{(Z_\ell + \mu k_w)}_{Z_{eff}^{Sat}} \qquad \theta_{eff}^{Sat} = \tan^{-1}\left(\frac{X}{R+\mu k_w}\right) \xrightarrow[\mu\uparrow]{} 0$$

Key conclusion: A real k_w adds only resistance: no independent angle-shaping freedom

Solution Using Hybrid Current Limiter: SatLim with Virtual Anti-windup

Key idea

- Replace the real gain k_w with VI
- The hybrid limiter separates current-magnitude and current-angle control
- Direct saturation enforces the hard current limit, while VI steers the limited-current direction



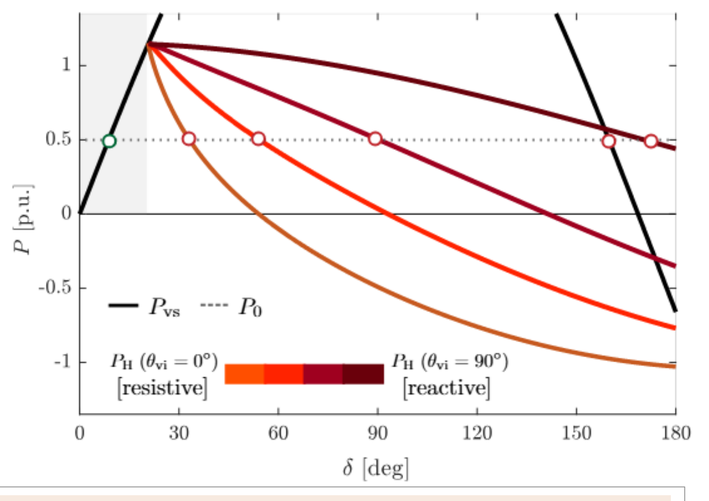
Hybrid limiter: SatLim with virtual anti-windup

From network: $E - V_g = IZ_\ell$ GFM control: $E = E^\star - (R_{vi} + jX_{vi})(I^\star - I)$

$$E^\star - V_g = I\underbrace{[Z_\ell + \mu(R_{vi} + jX_{vi})]}_{Z_{eff}^{H}} \qquad \theta_{eff}^{H} = \tan^{-1}\left(\frac{X+\mu X_{vi}}{R+\mu R_{vi}}\right) \xrightarrow[\mu\uparrow]{} \tan^{-1}\left(\frac{X_{vi}}{R_{vi}}\right) = \theta_{vi}$$

The new degree of freedom is the angle of the added VI term: it can shape the current angle

With increasing μ, the effective angle rotates toward the selected θ_{vi}



Angle-shaping ability: $\mu \to 0: \ \theta_{eff}^{Sat} \to \theta_\ell, \ \theta_{eff}^{H} \to \theta_\ell,$
Scalar vs. virtual AW $\mu \to \infty: \ \theta_{eff}^{Sat} \to 0, \ \theta_{eff}^{H} \to \theta_{vi}.$

Trade-off: high limited power, weak recovery slope

$$P_H^2 + 4K^2 = (|V_g| I_{lim})^2, \qquad K = -\frac{dP_H}{d\delta} > 0$$

Key conclusion

Unstable equilibrium point shifts towards right: phase jump boundary increases

Ride-through secured but at a price: the flat branch has weak restoring stiffness, so recovery is slow

Recovery Time Reduction Using Hybrid Limiting & Added Compensations

Fast recovery requires a larger restoring power margin

After a positive phase jump, the angle must decrease to exit CL mode. A larger difference between perceived and commanded power makes it return faster

From P-f droop law, $\dot\delta = -m_p\underbrace{(P_{per} - P_{cmd})}_{\text{restoring power margin}}, \qquad P_{per} = P_H + P_f, \qquad P_{cmd} = P^\star - \Delta P^\star$

S_1 - Fictitious-power compensation

$P_{per} = P_H + \mu\Re\{EI^*\} \approx P_{VS} \uparrow$

Raises perceived power from withheld anti-windup voltage

S_2 - AW-voltage power feedback

$P_{per} = P_H + kv_{aw,q} \uparrow$

Converts q-axis withheld voltage into the added power term

S_3 - Angle-power feedback

$P_{per} = P_H(\delta) + \frac{k}{m_p}\chi(\rho(\delta))\delta \uparrow$

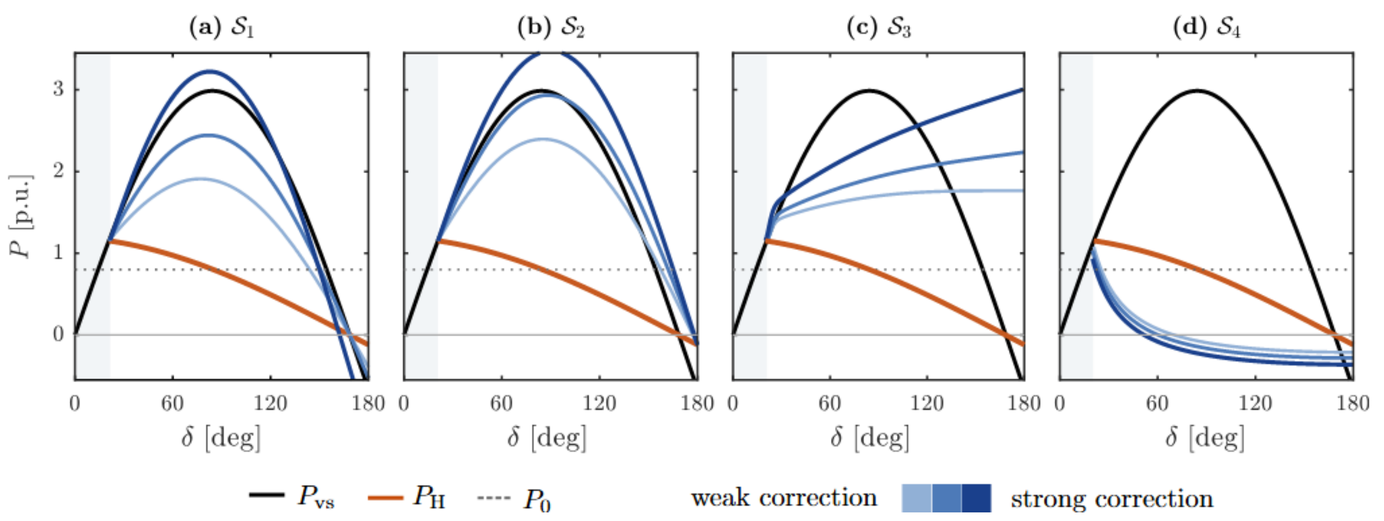
Converts angle error into an equivalent added active power term

S_4 - Dynamic set-point reduction

$P_{cmd} = P^\star - (k_p^d(\varepsilon + \varepsilon_b s_{lim}) + k_i^d\xi) \downarrow$

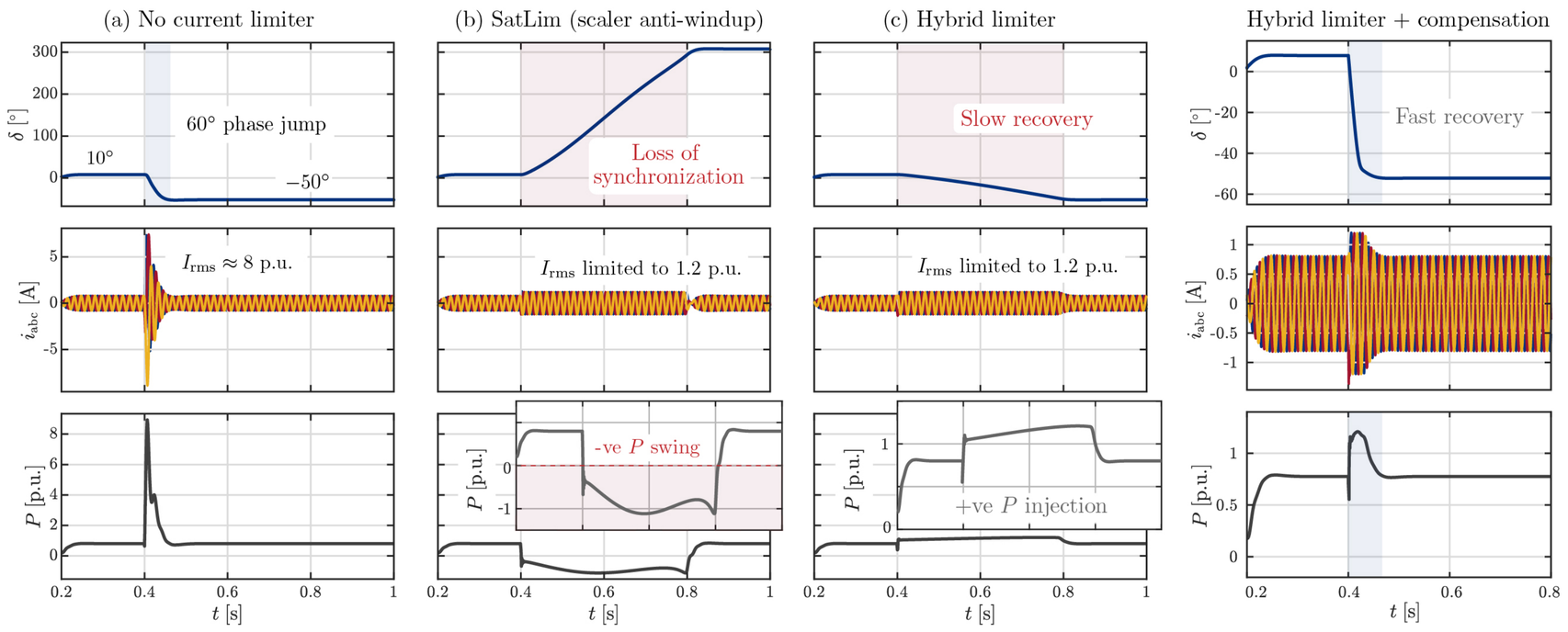
PI regulator decreases power setpoint during phase jump event

Goal: enlarge (P_{per} - P_{cmd}) until limiter exit; every compensation is then reset



$\mathcal{S}_1 - \mathcal{S}_3$ raise the active power perceived by the synchronization loop to restore a positive synchronizing slope
$\mathcal{S}_4$ temporarily reduces the commanded power toward the achievable hybrid-limited power

Simulation Results



Conclusion

- Under saturation, saturation-based limiter can lead to loss of synchronization and negative active power swing
- Hybrid current limiter (SatLim + virtual-impedance anti-windup) removes the dip and preserves synchronism at the same hard current limit
- Limiter-gated compensations can restore voltage-source recovery speed

Acknowledgements

U.S. DEPARTMENT of ENERGY

unifi consortium

[1] D. Chatterjee, N. Baeckeland, B. Johnson, and Gab-Su Seo, "Phase Jump Ride-through of Current-limited Grid-forming (GFM) Inverters," IEEE TPEL (in preparation)